ROBO RUN — Game Design Document

Game Development Fundamentals · Assignment 8–9 · Author: Lunara · Engine: Godot 4.7 (Mobile renderer), GDScript

1. Title + pitch

Robo Run. A little robot runs and jumps across floating sky islands, collects every coin to open the exit portal, and finally knocks down the Stone King.

2. Genre · platform

3D action platformer. Android first (iOS possible later, same project). Landscape only (sensor landscape), reference resolution 1920 × 1080.

3. 2D or 3D

3D, third-person camera. The hero is a real rigged character (RobotExpressive, Quaternius, CC0) with an AnimationTree: Idle / Run / Jump, Dance on the menu and win screens.

4. Core loop

Run → Collect coins → Open the portal → Reach the next island → (repeat, harder). Enemies and traps sit between the coins, so the player also has to dodge or shoot.

5. Controls (on-screen, thumbs in the bottom corners)

Control	Where	What it does
Virtual stick	bottom-left	move the robot (camera-relative)
JUMP	bottom-right	jump (also onto moving platforms)
FIRE	bottom-right, above JUMP	shoot a bullet forward (cooldown)
II (pause)	top-right	pause: time stops; Resume / Restart / Main menu
Drag on free screen area	anywhere else	rotate the camera

6. Content

Levels	Enemies / hazards	Items
Now: 3 (L1 tutorial islands, L2 dangerous path, L3 boss arena). Final: 4 levels + boss.	Patrol bomb, falling rocks, spike traps, Stone King boss (fireballs, rock rain). Planned: 1 chasing enemy.	Coins (score; all needed to open the portal), heart pickup (+1 HP). Hero has 3 hearts.

7. Art and sound

Bright low-poly pastel world: green islands, sky gradient, one soft sun. Models: Quaternius (robot, CC0) and Sketchfab CC-BY models (rock, bomb, heart, spike trap), recoloured and lit the same way so it reads as one world. UI: one dark rounded theme with white text and red hearts, same on every screen. Sound: SFX synthesised by me (jump, coin, hurt, shoot, portal, boss); one looping music track (source and licence to be credited in README).

8. MVP — must be in the week-15 build

Main menu, HUD, pause and win screens that fit every phone and tablet · 3 levels + boss fight · touch controls · enemy AI (patrol / chase / attack, boss phases) · saving (unlocked level, best coins, volume) · music + SFX with volume settings · hit / coin / portal effects · installable Android APK.

9. Cut list — decided NOT to make

Online leaderboard · shop, skins, currency · several playable characters · procedural levels · cutscenes / story dialogues · in-app purchases and ads · portrait mode · multiplayer.

10. Plan, weeks 9–13 (then teams in week 14, final build in week 15)

Week	Ready at the end of the week
9	This GDD; GitHub repository with clean history; Menu / HUD / Pause screens with stretch scaling, anchors and SafeArea, tested on 16:9, notch phone and 4:3 tablet.
10	Enemy AI: patrol bomb as a state machine (patrol → chase → attack), Stone King phases; one new chasing enemy.
11	Saving: unlocked levels, best coins per level and settings in user:// (ConfigFile); Continue button in the menu.
12	Sound: music + SFX buses, volume sliders in a Settings screen, sounds for all actions.
13	Effects and polish: particles for coins, hits and portal, camera shake, level 4; Android build tested on a real phone.

Roles

Solo until week 14: Lunara — game design, programming, UI, level design, sound, integration and Git. When teams form, the roles will be split (programming / level and art / UI and sound).